ECN 594: Oligopoly Refresher

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Plan

1. **Cournot competition (refresher)**
2. Bertrand competition (refresher)
3. Escaping the Bertrand paradox

From ECN 532: Oligopoly models

- You covered Cournot and Bertrand in Hector's class
- Today: quick refresher in IO notation
- **New focus:** Market power measurement
    - Connecting oligopoly models to Lerner index
    - When does each model apply?

Cournot competition: setup

- n firms producing homogeneous goods
- Firms choose **quantities** simultaneously
- Inverse demand: $P = P(Q)$ where $Q = \sum_{i=1}^{n} q_i$
- Constant marginal cost: c
- Firm i profit: $\pi_i = P(Q) \cdot q_i - c \cdot q_i$

Worked example: Cournot duopoly

- **Question:** Inverse demand is $P = 100 - Q$. Two symmetric firms with $MC = 10$.
- Find equilibrium quantities and price.

Worked example: Cournot duopoly (solution)

- **Solution:**
- Firm i profit: $\pi_i = (100 - q_i - q_j)q_i - 10q_i$
- FOC: $\frac{d\pi_i}{dq_i} = 100 - 2q_i - q_j - 10 = 0$
- Symmetric: $q_1 = q_2 = q^*$, so $100 - 3q^* = 10$
- $q^* = 30$
- $Q = 60$, $P = 100 - 60 = 40$
- Profit per firm: $\pi = (40 - 10) \times 30 = 900$

Cournot with n symmetric firms

- **Setup:**
- n firms with constant marginal cost c
- Linear demand: $P = a - bQ$ where $Q = q_1 + q_2 + \ldots + q_n$
- **Question:** What are the Cournot equilibrium quantities and price?

Cournot with n symmetric firms (solution)

- Firm i profit: $\pi_i = q_i(a - bQ) - cq_i$
- FOC: $\frac{d\pi_i}{dq_i} = a - bQ - bq_i - c = 0$
- By symmetry: $q_1 = q_2 = \ldots = q_n = q^*$, so $Q = nq^*$
- Substituting: $a - bnq^* - bq^* - c = 0$
- Solving: $q^* = \frac{a-c}{b(n+1)}, \quad P^* = \frac{a+nc}{n+1}$
- **Key insight:** As $n \to \infty$, $P \to c$ (perfect competition)

Practice: Cournot comparative statics

- **True, False, or NEI:**
- (a) Adding a third firm to a Cournot duopoly always reduces industry profits.
- (b) In Cournot, all firms must have the same marginal cost.

Practice: Cournot comparative statics (solution)

- **Answers:**
- **(a) TRUE.** More firms $\Rightarrow$ lower price $\Rightarrow$ lower industry profit. Each firm's profit falls, and total profit falls because P is closer to MC.
- **(b) FALSE.** Asymmetric costs work fine. Low-cost firms produce more and earn higher margins.

Plan

1. Cournot competition (refresher)
2. **Bertrand competition (refresher)**
3. Escaping the Bertrand paradox

Bertrand competition: setup

- n firms producing **homogeneous** goods
- Firms choose **prices** simultaneously
- Consumers buy from lowest-price firm
- If tie: split demand equally
- Constant marginal cost: c

Bertrand: the paradox

- **Nash equilibrium:** $p_1 = p_2 = c$ (marginal cost pricing!)
- **Why?**
  - If $p_i > p_j > c$: firm i can undercut and capture entire market
  - Undercutting continues until $p = c$
- **The "paradox":**
  - Only 2 firms, but competitive outcome!
  - Zero profits with just 2 competitors
  - Seems unrealistic for most markets

Potential solutions to the Bertrand paradox

- How could we change the assumptions of the Bertrand model to get a more realistic model with positive profits (i.e. firms pricing above MC)?

- **Asymmetric (i.e. different) costs**
  - Like 'cost leadership'

- **Product differentiation/branding**
  - Undercutting the price by a small amount may no longer deliver all of total demand

- **Dynamic competition**
  - What if rivals can retaliate? E.g. you set a high price but threaten to retaliate next time if your rival undercuts you?

- **Capacity constraints**
  - What good is undercutting your rival to get total demand if you cannot actually supply all this demand due to capacity constraints?

Escaping the Bertrand paradox

1. **Capacity constraints** (Kreps-Scheinkman)
   - Can't serve entire market at low price
   - Leads to Cournot outcome
2. **Product differentiation** (next lecture!)
   - Consumers have preferences
   - Not all switch to lowest price
3. **Repeated interaction** (collusion, covered later)
   - Firms can sustain $P > MC$ through punishment strategies

When does each model apply?

- **Cournot applies when:**
    - Capacity constraints matter
    - Firms commit to production before selling
    - Quantities are hard to adjust quickly
- Examples: manufacturing, airlines (seat capacity)
- **Bertrand applies when:**
    - Prices adjust quickly
    - No capacity constraints
    - Homogeneous products
- Examples: online retail, commodities

Industry examples: which model?

Industry	Model	Why?
Airlines	Cournot	Capacity (planes, gates) committed
Cement	Cournot	Production committed; transport costs
Gasoline stations	Bertrand	Prices adjust daily; commodity
Online retail	Bertrand	Instant price changes; no capacity
Smartphones	Differentiated	Product differentiation dominates

- In practice: most markets have differentiated products!

Practice: Cournot vs Bertrand

- **Question:** Two firms have $MC = 20$. Market demand is $P = 100 - Q$.
- (a) Find equilibrium price under Cournot.
- (b) Find equilibrium price under Bertrand.
- (c) Which model yields higher consumer surplus? Why?

Practice: Cournot vs Bertrand (solution)

- **Solution:**
- **(a) Cournot:** Using $P^* = \frac{a+nc}{n+1} = \frac{100+2(20)}{3} = 46.67$
- Or: $q^* = \frac{100-20}{3} = 26.67$, $Q = 53.33$, $P = 46.67$
- **(b) Bertrand:** $P = MC = 20$
- **(c)** Bertrand has higher CS:
  - Bertrand: $CS = \frac{1}{2}(100-20)(80) = 3200$
  - Cournot: $CS = \frac{1}{2}(100-46.67)(53.33) \approx 1422$
- Lower price $\Rightarrow$ higher quantity $\Rightarrow$ higher CS

Kreps-Scheinkman (1983): resolving the puzzle

- Two-stage game:
  1. Stage 1: Firms choose capacities (quantities)
  2. Stage 2: Firms compete in prices
- **Result:** Equilibrium outcome = Cournot!
- **Intuition:**
  - Capacity choice commits firms
  - Price competition is constrained by capacity
  - Undercutting is limited by what you can produce
- Key insight: commitment matters

Oligopoly refresher: Key Points

1. **Cournot:** Firms choose quantities; $P > MC$
2. **Bertrand (homogeneous):** $P = MC$, zero profits
3. **Bertrand paradox:** Only 2 firms but competitive outcome
4. Cournot applies with capacity constraints; Bertrand with flexible prices
5. **Escaping the paradox:** Capacity constraints, differentiation, collusion
6. **Kreps-Scheinkman:** Capacity then price $\Rightarrow$ Cournot outcome

Next time

- **Lecture 8:** Product Differentiation and Hotelling
  - Product differentiation: why it matters
  - Hotelling model of spatial competition
  - Connection to demand estimation